

PUMP P-101 ? CENTRIFUGAL FEED PUMP

MAINTENANCE & OPERATIONS MANUAL Rev 4 | Jan 2025

Equipment Tag : P-101 (Spare: P-101S)

Location : Unit 2 ? Coker Feed Section, Platform Level EL+12.0 m

Manufacturer : KSB India Ltd (Model: Etanorm SYT 150-315)

Rated Flow : 450 m³/h Rated Head : 120 m NPSH required: 4.2 m

Motor : 250 kW, 2-pole, FLP enclosure, IE3 efficiency class

Fluid : Coker feed (Vacuum residue), SG 0.92, 185 °C at inlet

1. OVERVIEW

P-101 delivers heated feedstock from the surge drum D-204 to fired heater H-301.

It is a critical-service pump operating 24×7. Standby P-101S must remain on auto-standby and should be alternated every 72 hours per the rotating equipment policy. Any trip of P-101 automatically starts P-101S within 8 seconds.

2. ROUTINE MAINTENANCE SCHEDULE

Daily checks (operator, each shift):

- ? Seal flush flow indicator: minimum 3 L/min; low flow triggers alarm PA-2101
- ? Bearing temperature: alarm 85 °C, trip 95 °C (bearing RTDs TI-2101A/B)
- ? Vibration: alarm 7.1 mm/s, trip 11.2 mm/s (transmitters VT-2101A/B)
- ? Suction pressure: minimum 2.5 barg at PI-2101; below 2.1 barg causes cavitation

Weekly (area technician):

- ? Inspect mechanical seal for weeping; any visible drip > 5 drops/min ? WO request
- ? Record suction (PI-2101) and discharge (PI-2102) pressures in log sheet FM-M-014

Monthly:

- ? Grease motor bearings: Kluber Isoflex NBU15 grease, 3 shots per side; do NOT over-grease
- ? Check coupling alignment; maximum allowable: 0.10 mm parallel, 0.05 mm angular

Annual Overhaul:

- ? Pull pump for full teardown; replace mechanical seal (Item 433), wear rings, and bearings
- ? Laser-align coupling within 0.05 mm
- ? Hydraulic test casing at 1.5× operating pressure

Known failure mode ? Cavitation-induced seal failure:

Repeated seal failures on P-101 (WO#8821, WO#9034, WO#9412) have been traced to cavitation when D-204 level falls below 40%. Always confirm drum level > 40% before ramping throughput above 380 m³/h. Install low-level interlock LI-2204 per RCA#2024-07.

3. SAFETY PRECAUTIONS

P-101 handles hydrocarbon at 185 °C, well above auto-ignition temperature (AIT ~250 °C).

Pre-Maintenance Mandatory Steps:

1. Obtain Maintenance Work Permit (MWP) signed by Area Authority and Safety Officer
2. Lock out and tag out (LOTO) motor at MCC panel ? minimum 1 padlock per worker
3. Isolate suction (XV-2101) and discharge (XV-2102) block valves; install spades
4. Depressurise to closed blowdown header (BD-03); verify zero pressure at PI-2101
5. Drain and flush casing; confirm skin temp < 40 °C before touching
6. For pump pit access: gas test LEL < 10%, O₂ 19.5?23.5%, H₂S < 10 ppm
7. Issue Confined Space Entry Permit (CSEP) ? do NOT enter without valid CSEP

Hot-Work Restriction:

No hot-work permit (HWP) may be active within 15 metres of P-101 during any maintenance involving casing opening. SIMOPS (Simultaneous Operations) Procedure OISD-GDN-192 applies. Inform shift supervisor before issuing HWP in adjacent area.

Do NOT dead-head the pump (run against closed discharge valve) for more than 30 seconds ? thermal runaway occurs within 90 seconds and can rupture the seal.

4. TROUBLESHOOTING GUIDE

Symptom: High vibration (> 7.1 mm/s)

Possible causes:

- a. Cavitation ? check suction pressure; confirm D-204 level > 40%
- b. Bearing wear ? pull bearing for inspection; replace if pitting or spalling
- c. Coupling misalignment ? laser-align; tolerance ? 0.10 mm

Symptom: Seal leakage

Possible causes:

- a. Cavitation damage to seal faces ? inspect and replace seal
- b. Flush flow inadequate ? clean strainer in flush line; min 3 L/min
- c. Shaft run-out > 0.05 mm ? check shaft straightness; replace if bent

Symptom: Low discharge pressure

Possible causes:

- a. Wear rings eroded ? measure clearance; replace if > 1.5× OEM spec
- b. Suction strainer fouled ? schedule clean-out on next shutdown
- c. Wrong rotation direction ? verify after any motor rewind

5. SPARE PARTS LIST (CRITICAL SPARES ? MAINTAIN ON-SITE)

Item 433 Mechanical seal (EagleBurgmann BRT-D) ? 2 sets

Item 321 Radial bearing (SKF 6215) ? 4 units

Item 325 Thrust bearing (SKF 7315) ? 2 units

Item 201 Wear ring set ? 1 set

Item 501 Coupling element (Rexnord M90) ? 2 units